SHEET # 1

1.1 Define the following items: Signal, system, signal processing. (State one example for each).

1.2 Classify the different types of signal processing.

1.3 Classify the different signal types.

1.4 Explain the basic three operations performed in A/D converters.

1.5 State and explain the sampling theorem.

1.6 Determine which of the following sinusoid signals are periodic, and then compute their fundamental period.

(a) $x_{a(t)}$ = 3cos(5t +π/6) (b) $x_{a(t)}$ =3cos(100πt)

(c) x(n) = cos[π(30n/105)] (d) x(n) = 2exp.[j(n/6-π)]

1.7 Consider the following analog sinusoid signal: $x_{a(t)}$= 3sin (100πt)

(a) Sketch the signal for [0, 30] ms.

(b) The signal $x_{a(t)}$ is sampled with sampling rate F_S=300 samples per second.

Determine the frequency of the discrete-time signal and show that it is periodic.

(c) Compute the sample values in one period of x (n). Sketch x (n) on the same diagram with

$x_{a(t)}$. What is the period of this discrete-time signal, x (n).?

(d) Can you find a sampling rate F_S such that the signal X(n) reaches its peak value 3 ? What is the minimum F_S suitable for this task?

1.8 Consider a simple signal processing system as shown in the following figure. The sampling periods of the A/D and D/A converters are T_1=5 ms and T_2=1 ms respectively. Determine the output y_a (t) if the input is given by:

$$x_{a(t)} = 3\cos(100\pi t) + 2\sin(250\pi t) \quad \text{(t in second)}$$

